Inside the mind of AI

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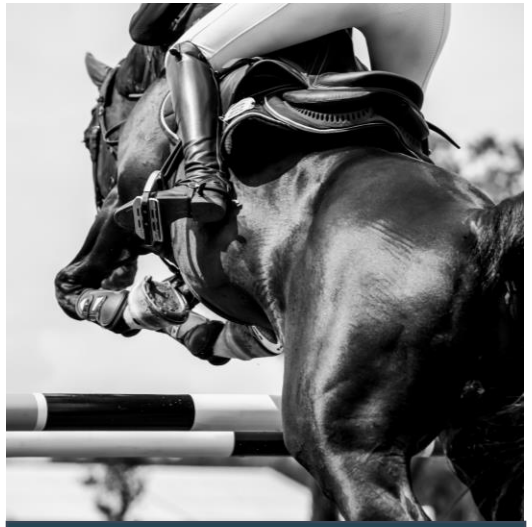
Educational session 6.3, 21/10/2026

MSToronto2026

10th Joint ACTRIMS-ECTRIMS Meeting

OCTOBER 21-23, 2026 | TORONTO, CANADA

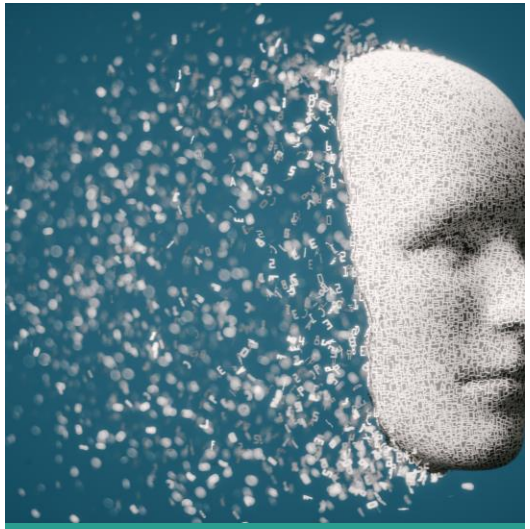
Educational objectives



Barriers

Clinical validation gap

1. Awareness



Inside the mind of AI

Beyond performance
Trustworthy AI

2. Knowledge



Multidisciplinary

Stakeholder engagement

3. Participation

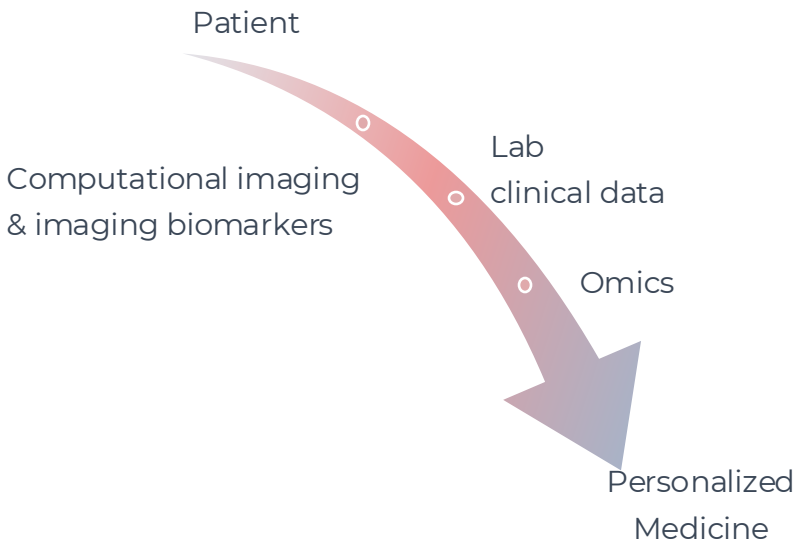
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AI in MS: clinical and research context

Diagnosis & Follow-up[1,2]: Disease modifying treatment ?

Disease mechanisms & trajectories[3,4]





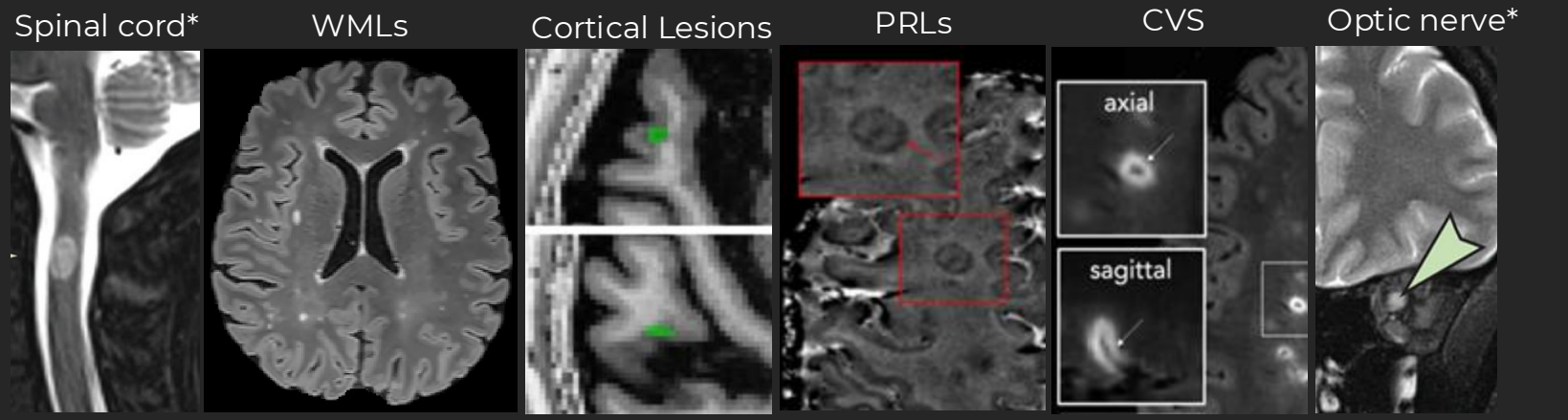
[1]X. Montalban et al, The Lancet Neurology 2025; [2]H. Vrenken et al, Neurology 2021; [3]A. Eshaghi et al Nature Communications 2021; [4]H. Ganjgahi et al, Nature Medicine, 2025.

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MR imaging biomarkers

Increased MRI data[1]:

- brain, spine, optic nerve
- quantitative image contrasts[2]
- new MRI biomarkers[3,4]
- follow up imaging



* Thumbnails adapted from 1.

- Improve trhough plane spinal cord imaging,
- Find new lesions in the brain,
- Identifying CVS lesions for differential diaganosis,
- Find phenotypes in large scale cohorts,
- Global brain atrophy or brain age

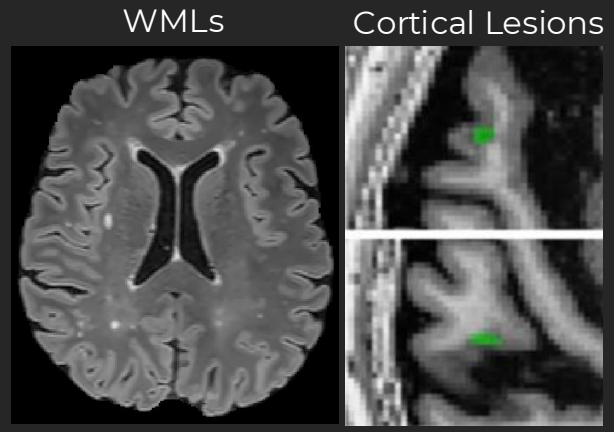
Wide range of machine learning tasks[5,6,7,8,9,10,11,12]

- Acquisition, reconstruction
- Image restoration, image synthesis
- Segmentation and classification
- Clustering and regression

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[1]F. Barkhof, The Lancet Neurology 2025; [2]C. Granziera et al, Brain 2021; [3] A. Cagol et al JAMA Neurology 2023; [4] H. Kolb et al, Neuroimage Clinical 2022; [5]F. La Rosa et al, Neuroimage Clinical, 2022; [6] R. Bonacchi et al, Neuroimage Clinical 2022; [7]S. Collorone et al, MS Journal 2024; [8] Hu et al Imaging Neuroscience 2025; [9] Manning et al AJNR Am J Neuroradiol 2025 ; [10] Bos et al Neuroimage Reports 2025; [11] JPR Falet et al Front Art Intell 2025 [12] Molchanova et al Neuroimage Clinical 2026.

MR imaging biomarkers



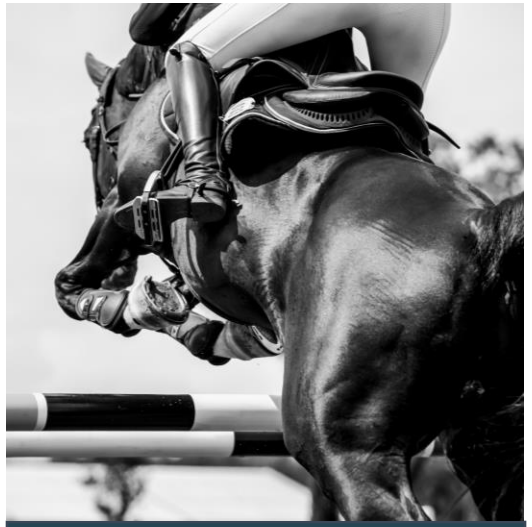
As MD, have you ever wanted / needed:

- Find new lesions in logitudinal follow up

- Segmentation

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OUTLINE



Barriers

The validation gap

The validation gap

development

Static (retrospective, often curated cohorts)

Mathematical / technical measures of accuracy

deployment

Dynamic (incomplete, quality, evolving)

Clinically relevant measures

Validation[1,2]



Technical : «engineer metrics», accuracy, AUC, Dice etc



Clinical : clinician-rated variables, RCT, prospective studies

In use : benefit to patients, time gain, socioeconomic impact

[1] O. Goodkin et al Br J Radiol 2019; [2] Z. Mendhelson et al Neuroradiology 2022.

The validation gap

MS lesion segmentation methods[1]

155 → 3 Clinical

1 In use

MS commercial tools[2]

10 → 4 Clinical

1 In use

FDA-authorized AI devices[3,4]

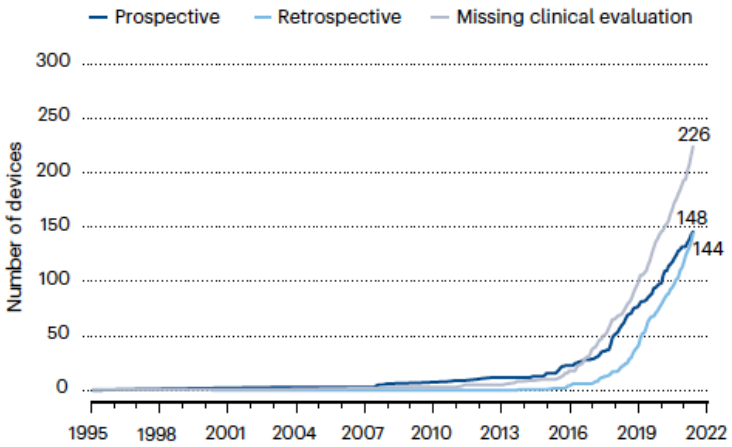


Figure adapted from Fassi et al Nature Medicine 2024

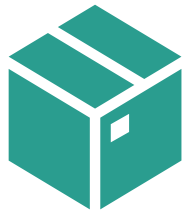


Figure adapted from Abulibdeh et al PLOS Digital Health 2026

[1] F. Spagnolo et al Neuroimage Clinical 2023; [2] Z. Mendhelson et al Neuroradiology 2022; [3] S. C. El Fassi et al Nature Medicine 2024; [4] R. Abulibdeh et al PLOS Digital Health 2026.

Barriers of AI adoption

It is about the model….



Validation gap, drift monitoring

It is about the data….



Infrastructure, governance

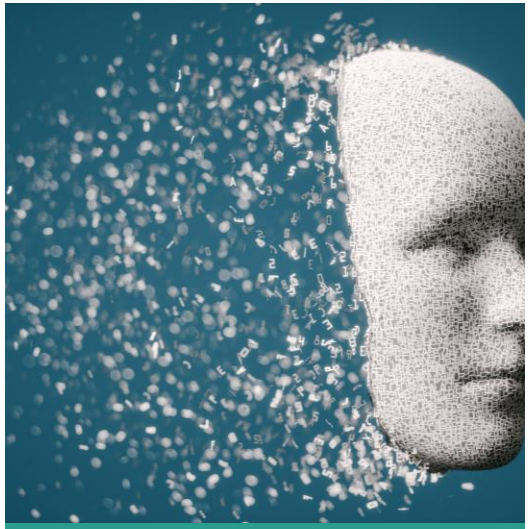
It is about the stakeholder….



Accountability, adoption of new technologies

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OUTLINE



Inside the mind of AI

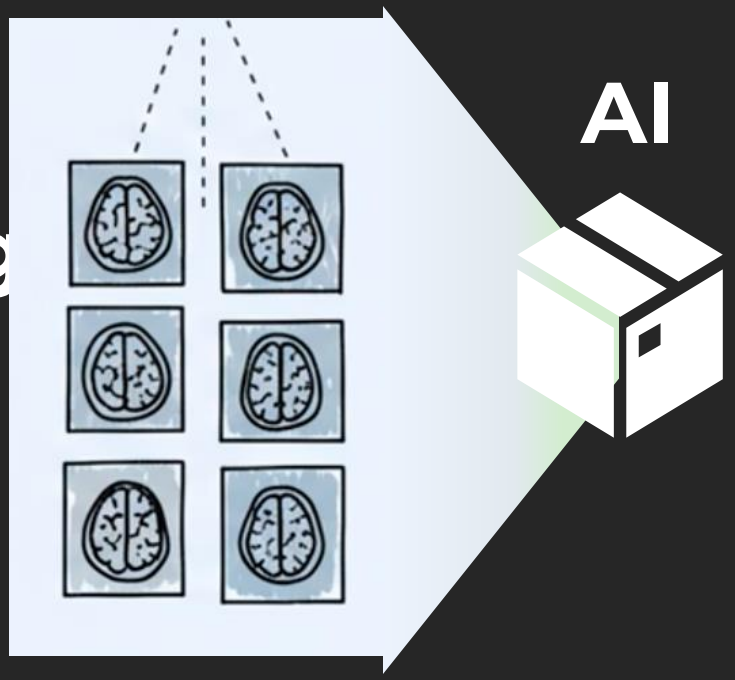
Beyond performance

INSIDE THE MIND OF AI

White matter and cortical lesion segmentation*

1. Training

MRI + annotations



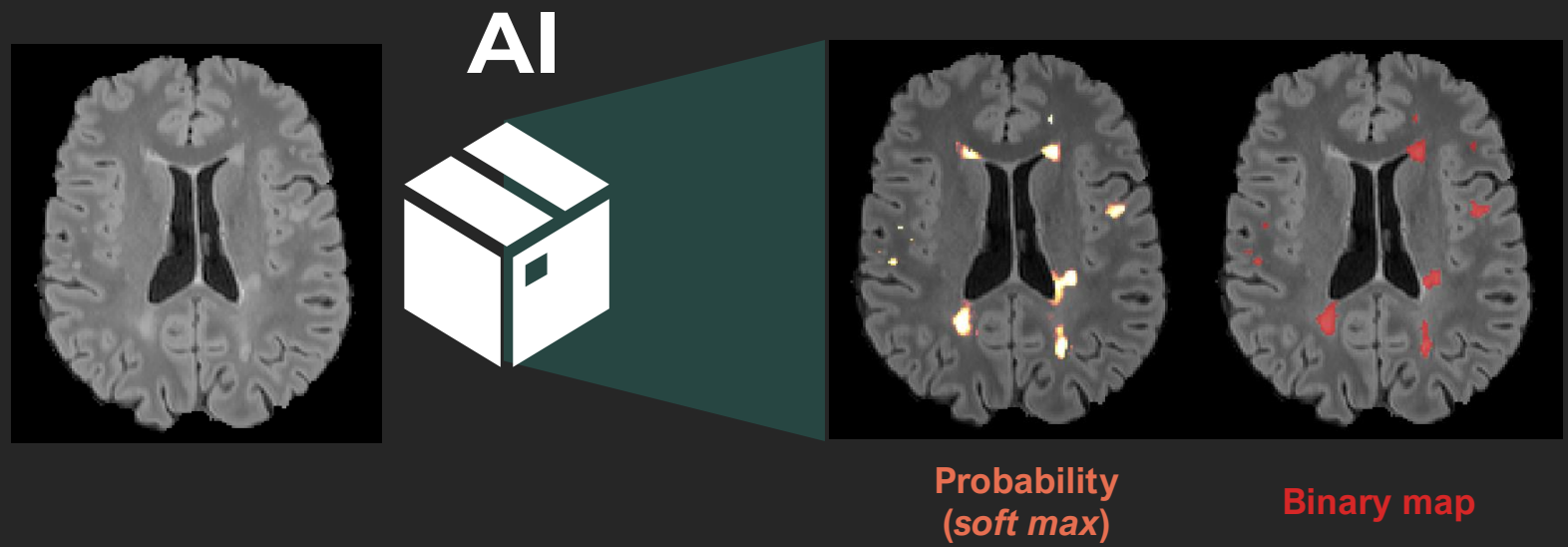
* Review papers : Commowick, O., et al Front. Neuroscience 2023, Diaz-Hurtado et al Neuroradiology 2022, La Rosa et al, Neuroimage Clinical, 2022; Molchanova et al Neuroimage Clinical 2026.

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White matter and cortical lesion segmentation*

2. Inference

Prediction

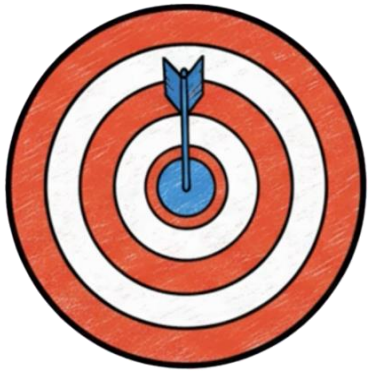


* Review papers : Commowick, O., et al Front. Neuroscience 2023, Diaz-Hurtado et al Neuroradiology 2022, La Rosa et al, Neuroimage Clinical, 2022; Molchanova et al Neuroimage Clinical 2026.

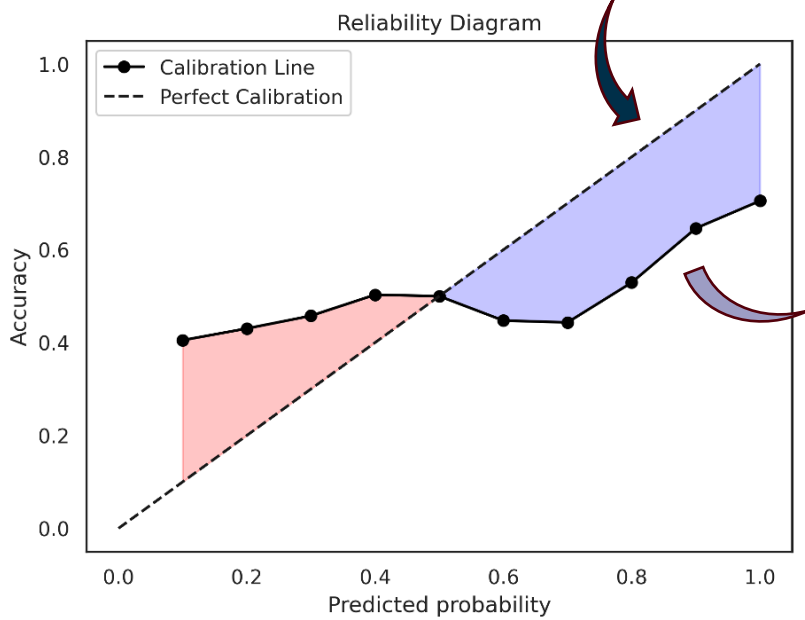
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Inside the mind of AI

Prediction: 90%



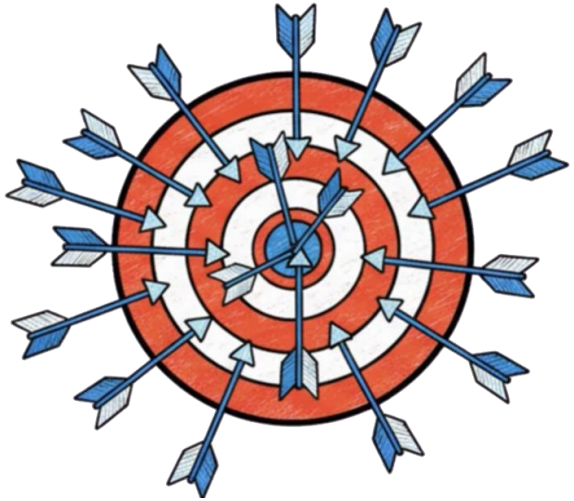
Calibrated model



Overconfident

Figure from Book Chapter by J. Deuschel et al 2024 in Unlocking Artificial Intelligence.

AI uncertainty quantification (UQ)[1,



Certainty of the prediction in MS lesion detection ?

Eg. Monte Carlo Dropout[6] or ensembling many decisions[7] (*frequentist* approach)

Variations are an indication of confidence of the model

[1]Abdar et al., Inf. Fusion, 2021. [2]Gal, Ph.D. Thesis, 2016. [3]Gawlikowski et al., AI Review, 2023. [4]Begoli et al., Nat. Mach. Intell., 2019; [5]B. Lambert et al, AI in Medicine 2024; [6] T. Nair et al Med Im Anal 2020; [7] N. Molchanova et al CIBM 2025.

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Inside the mind of AI: why AI models are uncertain

1 DATA NOISE

Image quality/artefacts

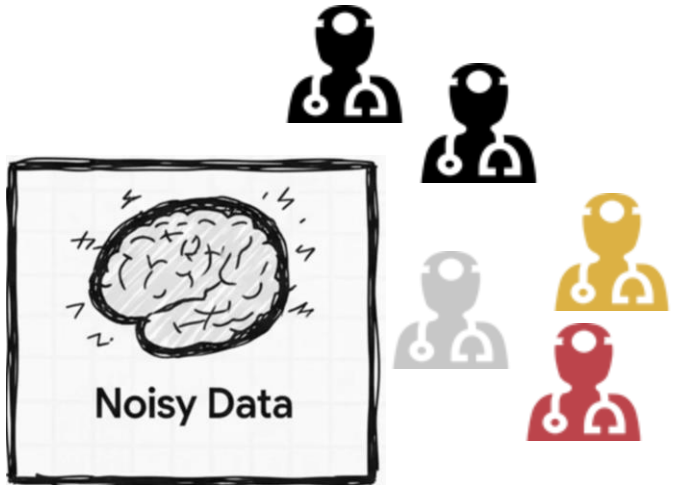
Annotations "noise"

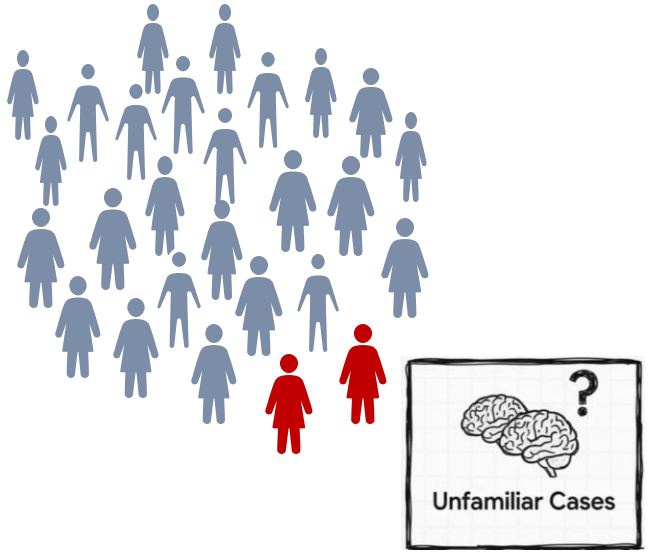
2 DATA SCARCITY

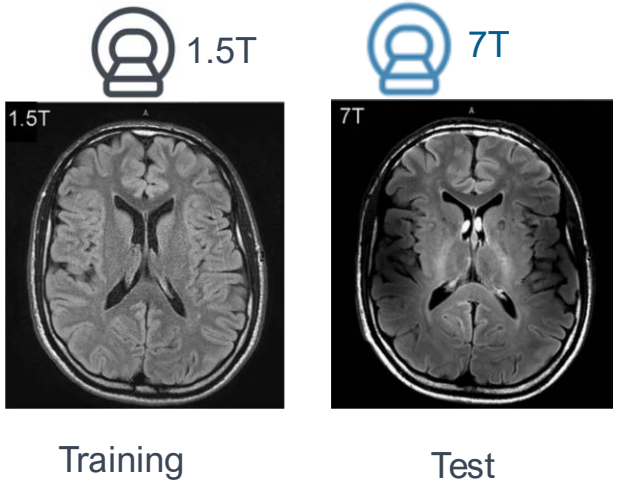
Few examples (hundreds) vs prevalence (millions)

3 DISTRIBUTIONAL SHIFTS

Data distributions in training and testing are different.







MSxplain

Responsible AI in Multiple Sclerosis

Explainability

Insight on internal actions and/or features of the DL-model.

Interpretability

Links the DL-model to the human observer knowledge.



Uncertainty

Quantification of the reliability of a model's prediction.

Integration

Trustworthy and transparency decision support for patient assessment.

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HINK Translational Imaging in Neurology

medGIFT

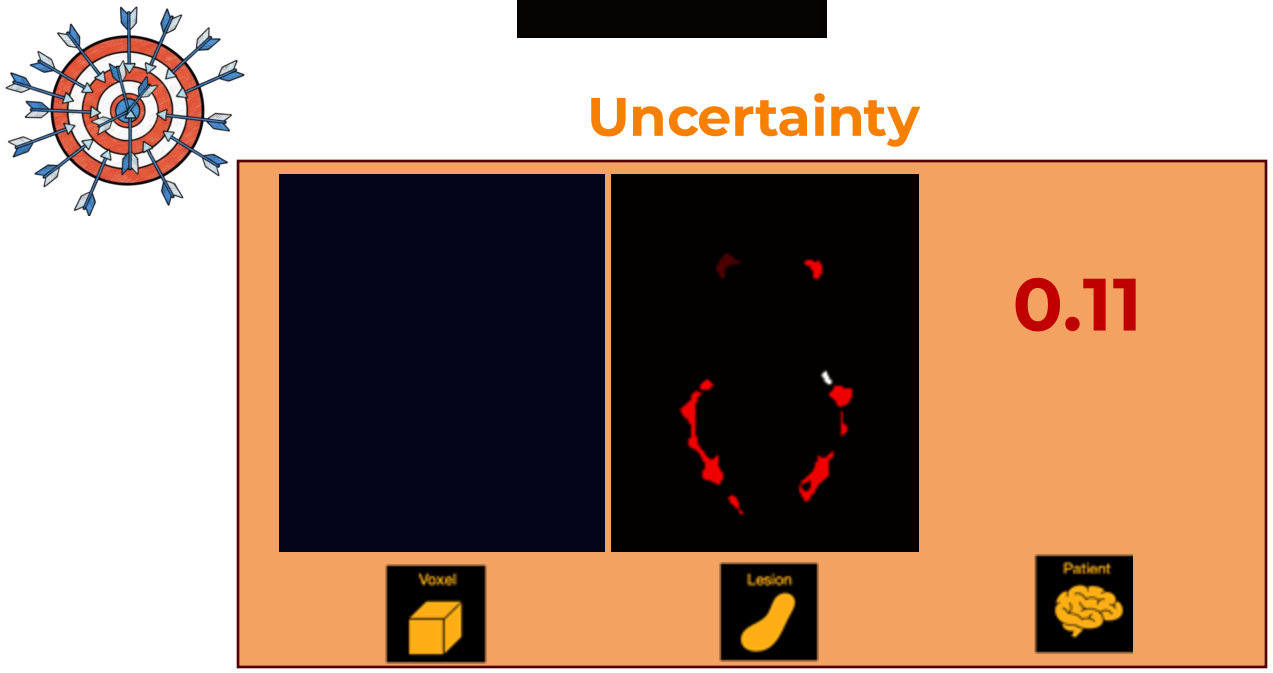
HASLERSTIFTUNG

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Inside the mind of AI: are you sure about that ?

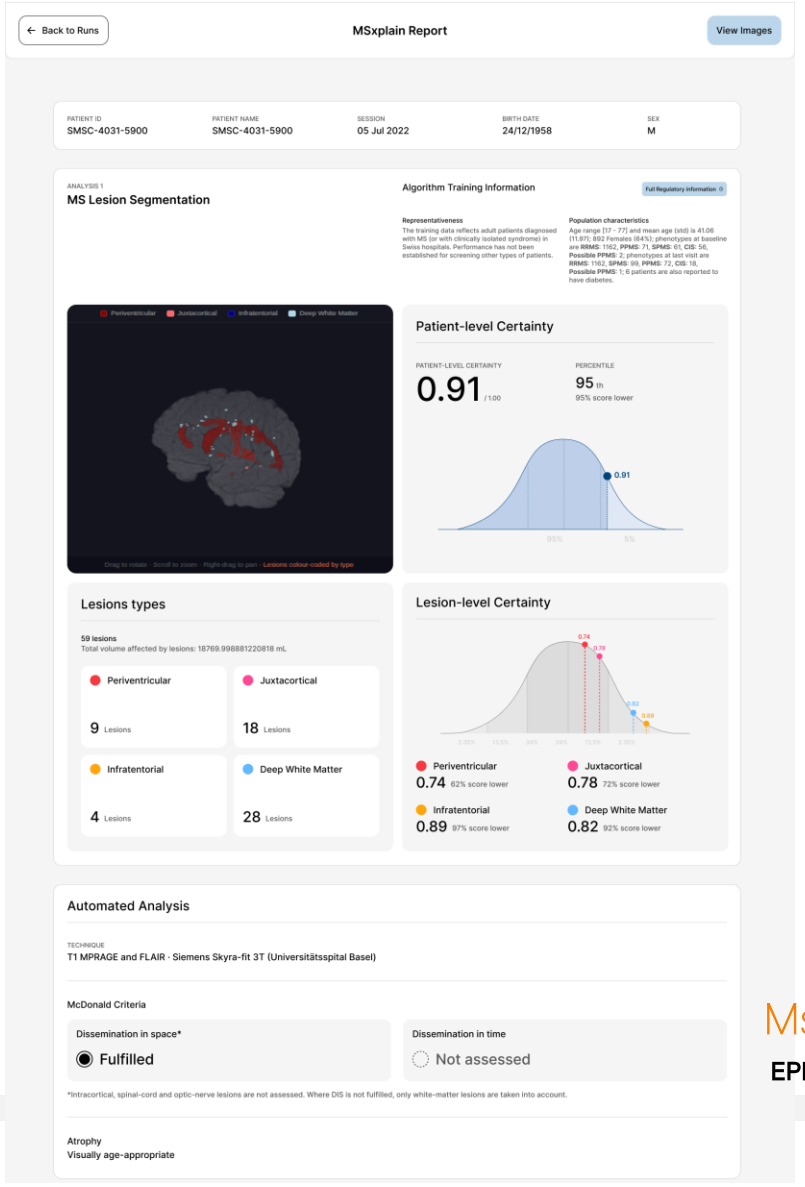
MSxplain

Where ? Uncertainty at different anatomical scales[1]



[1]N. Molchanova et al CIBM 2025.

How and when ? Uncertainty into the clinical wor



Msxplain + EPFL CALLAB

AI uncertainty in Multiple Sclerosis: what for ?[1]

Molchanova 2025 [2]

Lesion-scale uncertainty in segmentation is highly correlated with errors.

Useful in model drift monitoring.

Sreenivasan 2025 [3]

Electronic health records to classify to RRMS and SPMS. Conformal prediction model calibration.

Identified patients within the transition phase.

Storelli 2026 [4]

Prediction of cognitive worsening in MS, validation with UQ and explainability.

Support for flag cases to review.

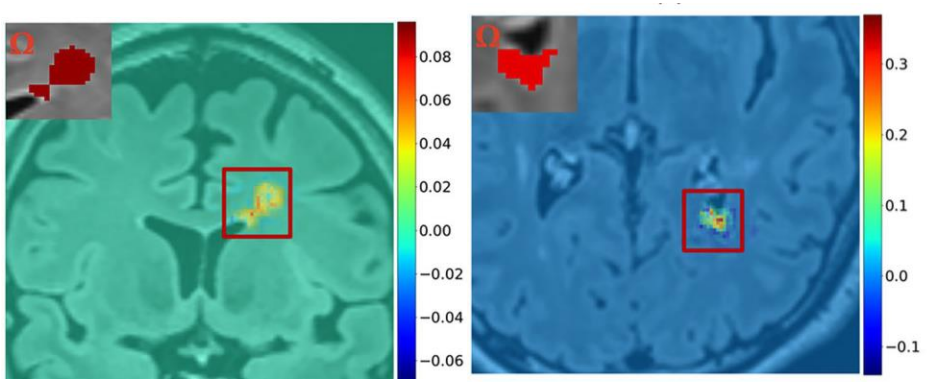
Uncertainty is inherent to medicine. It should be visible in medical AI.

1 CRS Banerji et al Nat. Med. 2023; [2] N. Molchanova et al CIBM 2025; [3] AP Sreenivasan et al npj Dig Med 2025; [4] L. Storelli et al Eu J. Neurology 2026.

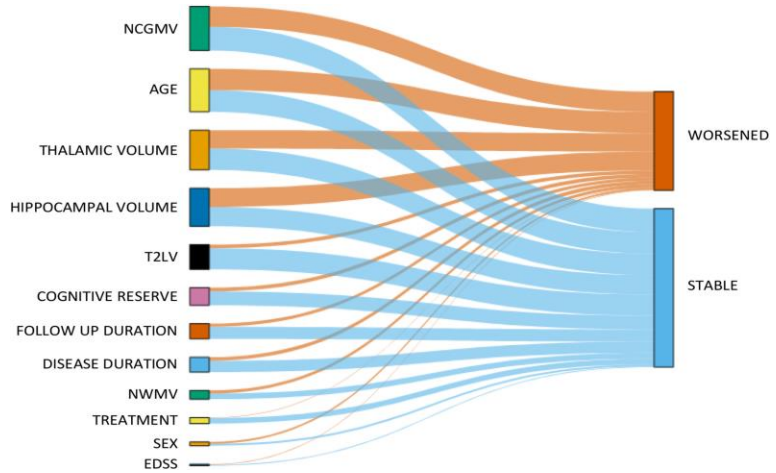
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Inside the mind of AI: explainable AI (XAI)

Which features guided AI decision?



F. Spagnolo et al Sci. Rep. 2026



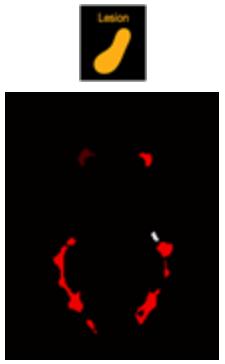
L. Storelli et al Euro J of Neurology, 2026

Why AI is uncertain ? [3]





X. Chen, A. Cagol (Univ. Basel), A Stölting (UCL)
E. S Beck (ISMMS), H. Tsagkas (NIH),





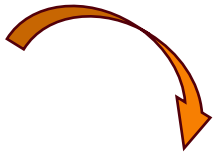


Image Quality
Lesion Location
Lesion Shape
Lesion Intensity
Demographics
...



N. Molchanova et al, arxiv. 2504.04814, 2025

Surveys of XAI : A. Saranaya et al Dec Anal J 2023, Haupt et al Eu J Rad AI 2026

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Trustworthy AI principles



Technical robustness

- Accuracy
- Resilience & safety
- Reliability, repeatability, reproducibility
- Uncertainty quantification & quality control

Human oversight

- Human AI - interaction
- Model drift monitoring
- Deployment, clinical validation

Accountability

- Life-cycle auditability
- Responsibility

Privacy

- Data Governance
- Data protection
- Data quality & integrity

Transparency

- Explainability
- Interpretaibitliy
- Traceability

Diversity

- Fairness
- Bias mitigation
- Inclusivity

Societal well-being

https://artificialintelligenceact.eu 2019 ; *FUTURE AI* , K. Lekadir et al BMJ 2025.
Figure from M. Zuluaga, I. Igsum, M. Bach Cuadra, Current Opinion in Biomedical Engineering 2026.

OUTLINE

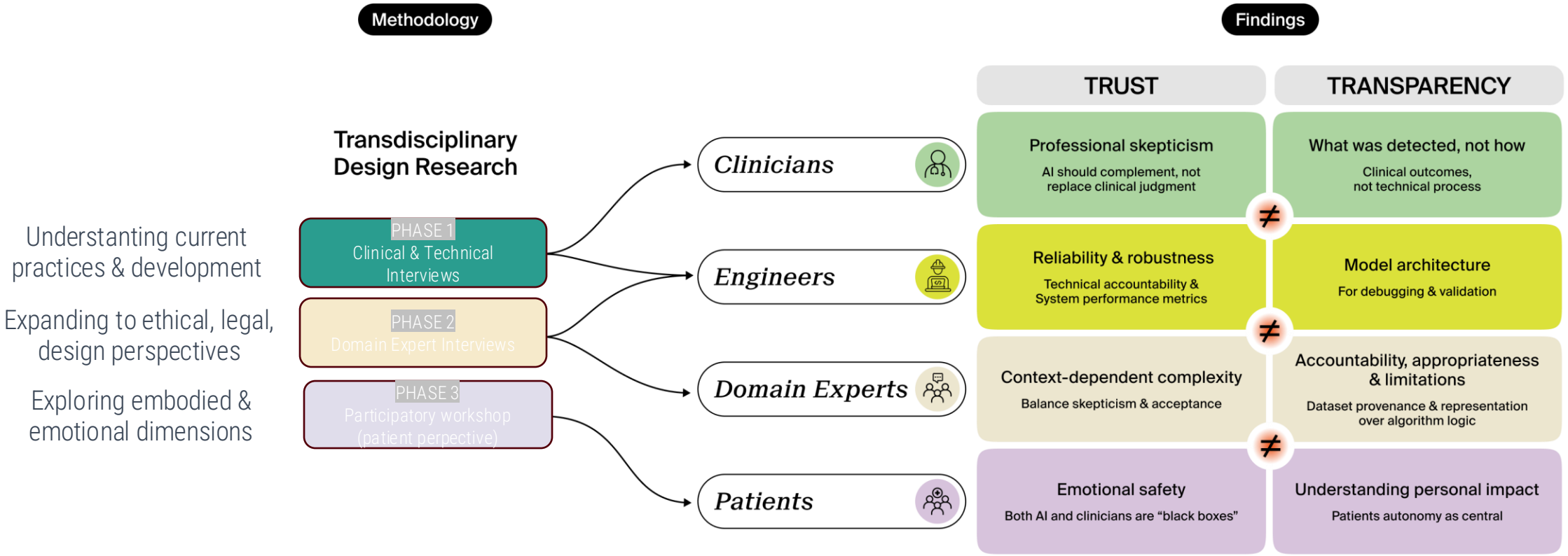


Multidisciplinary

Stakeholder engagement

Stakeholder-centered design

Understanding how diverse stakeholders perceive trust, transparency, and safety in XAI



Trust and transparency are conceptualized differently.
There are not universal solutions for implementing AI guidelines

Figure from M. Motta, Zenodo 10.5281/zenodo.18388638, Zhan et al IEEE CBMS 2025.

EPFL TECAL LAB

Trust and adoption of new technologies

Great, it performs super well !!



User is overconfident

Risk of automation bias

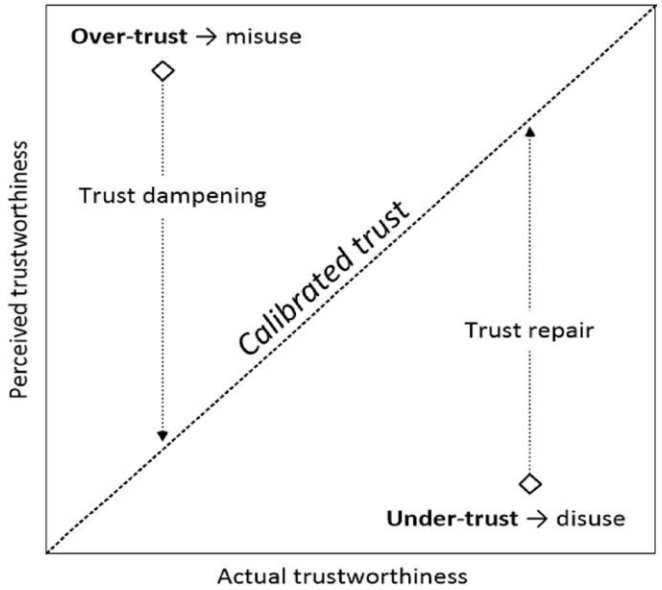


Figure from Visser et al, Int J. Social Robotics 2019

Wrong outpus initially



User is underconfident

Drops usage

O. Asan et al JMIR 2020, Mertz et al Front. Artif. Intell 2025

Multidisciplinary approach

MD, patients, industry, regulatory bodies, engineers

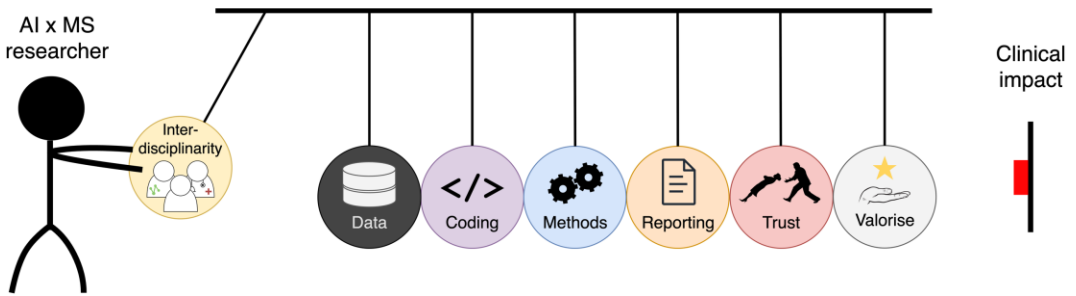
- Challenges ahead: different pace and time availability of stakeholders
- Quantify trustworthy AI principles
- Long-term and sustainable adoption of technology

Your feedback is needed !

Breaking barriers of adoption

Msxplain + EPFL ECAL LAB

iAIMS
international Community
for AI in Multiple Sclerosis



Denissen et al. 2026 –Under review

ECTRIMS
EUROPEAN COMMITTEE FOR TREATMENT AND RESEARCH IN MULTIPLE SCLEROSIS
AI and Digital Innovation Committee

actrims
AMERICAS COMMITTEE FOR TREATMENT AND RESEARCH IN MULTIPLE SCLEROSIS

Take home messages

Barriers for adoption

1. Awareness

- Validation
- Data
- Accountability

Inside the mind of AI

2. Knowledge

- AI uncertainty
- Explainability
- Interpretability

Multidisciplinarity

3. Participation

- Role of stakeholders
- Human-centric design
- MS community initiatives

Trustworthy AI principles & Calibrated Trust

Thanks to Patients

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Francesco La Rosa
Maxence Wynen
Germán Barquero
Hamza Kebiri
Nataliia Molchanova
Pedro Macias Gordaliza
Joe Najm

CHUV

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Patric Hagmann
Silvia Pistocchi
Théo Sartoretti

University of Basel

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Jannis Muller
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Muhamed Barakovic
Matthias Weigel
Po-Jui Lu
Matthias Mutke

UCLouvain

Pietro Maggi
Benoit Macq
Collin Vanden Bulcke
Anna Stölting
Serena Borrelli

Hes·so VALAIS WALLIS

Adrien Depeursange
Henning Muller
Mara Graziani
Lluis Borras
Roger Schaer
Vincent Andrearczyk

Isomorphic Laboratories

Andrey Malinin

HUG

Maria Isabel Vargas

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Delphine Ribes
When Zang
Clara Evans
Margeritha Motta
Sebastien Lago

NIH

Daniel S. Reich
Haris Tsagkas

HUMANITAS RESEARCH HOSPITAL

Martina Absinta

Icahn School of Medicine at Mount Sinai

Erin S. Beck

iAIMS international Community for AI in Multiple Sclerosis

Mar Barrantes Cepas
Tommy Broeders
Stijin Denissen

Marie Curie
ITN TRABIT

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Faculté de biologie et de médecine

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CHUV

THANK YOU FOR YOUR ATTENTION

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